Bill of Materials: Showcase (1 month)

REVA AICTE IDEA Lab Portal: everything needed to run a full showcase with all APIs connected (real Microsoft sign-in, Razorpay payments)

Ref: IDEALAB-BOM-2026-02
Date: 29 September 2026
Period: 1 month
Currency: INR

Minimum (laptop showcase)	Laptop + fixed public link	Recommended: showcase on AWS	Of which Claude Pro (1 month)
₹2,500	**₹3,350**	**₹4,878**	**₹2,500**

No.	Item	What it is for	Provider	Needed?	Rate	1 month
A. Development tool						
1	**Claude Pro**	AI assistant (Claude Code): fixes, changes and support up to and during the showcase	Anthropic	Yes	₹2,500 / month	₹2,500
Total A						**₹2,500**
B. APIs and accounts the site connects to (fill these in .env)						
2	**Microsoft sign-in (Entra ID app registration)**	Real 'Sign in with Microsoft' for @reva.edu.in students and staff; gives MS_TENANT_ID, MS_CLIENT_ID, MS_CLIENT_SECRET	Microsoft / REVA IT (REVA's Microsoft 365)	Yes (or use the built-in mock)	Free	₹0
3	**Razorpay: test mode**	Payment API keys for 3D-print checkout; test payments, no real money; no KYC needed	Razorpay	Yes	Free	₹0
4	**Razorpay: live mode (optional)**	Only if the showcase should take real money; needs business KYC	Razorpay	Optional	2% + 18% GST per payment	₹0
5	**Email sending**	Emergency-admin password-reset emails. For the showcase they can stay in the server log	Amazon SES (with AWS) / none	Optional	Free up to limits	₹0
6	**SSL certificate (HTTPS)**	Padlock / secure link; required by Microsoft for a public address	Let's Encrypt or tunnel	Yes (for public link)	Free	₹0
Total B						**₹0**
C. Where the site runs for the month: choose one						
7	**Option 1: your laptop + free tunnel**	Laptop is the server; free Cloudflare quick tunnel gives a temporary https link (changes each restart)	Cloudflare (free)	Choose one	Free	₹0
8	**Option 2: your laptop + fixed link**	Same, but a permanent https link, so the Microsoft redirect address never changes	ngrok Personal (approx.)	Choose one	~$10 / month	₹850

No.	Item	What it is for	Provider	Needed?	Rate	1 month
9	**Option 3: AWS server for the month (recommended)**	Runs 24/7 on the real server (EC2 t3.small, disk, IP, backups, email) with its own https address; also rehearses the final deployment	AWS Mumbai	Choose one	see full BoM	₹2,378
Option 3 amount includes 18% GST and matches section A of IDEALAB-BOM-2026-01. Only one option's cost applies.						
D. Already available (no cost)						
10	**Laptop, internet, Docker, Node.js, MongoDB**	Existing hardware and free open-source software	Existing / open source	Yes	Free	₹0
11	**Demo data and scripts**	Demo accounts, sample events, inventory and projects (npm run seed:demo)	In the repository	Yes	Free	₹0

Cost summary (1 month)

Showcase set-up	Claude Pro	Hosting / link	APIs	Total (1 month)
Option 1: laptop + free tunnel (minimum)	₹2,500	₹0	₹0	**₹2,500**
Option 2: laptop + fixed link	₹2,500	₹850	₹0	**₹3,350**
Option 3: AWS server (recommended)	₹2,500	₹2,378	₹0	**₹4,878**

API keys checklist: what to fill in .env

Setting in .env	Value comes from	Needed for
`PUBLIC_URL`	The site's address: http://localhost:3001, the tunnel https link, or the AWS domain	Everything
`JWT_SECRET`	Any 64 random characters (you make it)	Sign-in sessions
`MONGODB_URI`	mongodb://127.0.0.1:27017 on the laptop (Docker Compose sets it on AWS)	Database
`MS_TENANT_ID`	Azure portal → Entra ID → App registrations → the app → Directory (tenant) ID	Microsoft sign-in
`MS_CLIENT_ID`	Same page → Application (client) ID	Microsoft sign-in
`MS_CLIENT_SECRET`	App → Certificates & secrets → New client secret → copy the Value	Microsoft sign-in
`RAZORPAY_KEY_ID`	dashboard.razorpay.com → Test Mode → Account & Settings → API Keys (rzp_test_...)	Payments
`RAZORPAY_KEY_SECRET`	Shown once with the Key ID	Payments
`RAZORPAY_WEBHOOK_SECRET`	Razorpay → Webhooks → add <PUBLIC_URL>/api/payments/razorpay/webhook (optional; needs a public link)	Payment backup confirmation

Setting in .env	Value comes from	Needed for
`EMAIL_FROM, AWS_REGION`	SES-verified sender (AWS option only); leave empty on the laptop	Password-reset email

Notes

1. Claude Pro is budgeted for one month only (₹2,500, as provided).
2. Microsoft sign-in needs REVA IT to register the app in REVA's Microsoft 365 (free, about 15 minutes; steps in DEPLOYMENT.md section 3). They must add the redirect address <PUBLIC_URL>/api/auth/microsoft/callback. Until then, the built-in mock sign-in can be used for the showcase.
3. Microsoft only accepts https addresses (or localhost). To let the audience use their own phones you need Option 1, 2 or 3; Option 1's link changes on every restart, so the redirect address must be updated in Azure each time.
4. Razorpay test mode is enough for a showcase: payments look and work exactly like real ones (UPI success@razorpay) but no money moves. Live mode needs KYC and charges 2% + GST only on real payments.
5. AWS and ngrok amounts are estimates at an assumed 1 USD = ₹85; confirm current prices before paying. New AWS accounts may have free-tier credits that reduce Option 3.
6. Not included: staff time and lab hardware.

Prepared by	Reviewed by	Approved by
Name, signature and date	Name, signature and date	Name, signature and date